

Self-Learning: Address Decoding Techniques & 8259 Cascading -- MPCA

Module 4

Definition: Address Decoding Techniques define how system address lines are mapped to peripheral chip selects (Absolute vs Partial), while 8259 Cascading expands interrupt handling capacity from 8 to 64 priority channels.

1. Absolute vs Partial Address Decoding

ABSOLUTE ADDRESS DECODING:

- ALL address lines (A0 - A19) are fully connected to decoders and chip logic.
- Each memory/IO location has exactly ONE unique physical address.
- Eliminates "Shadow Addresses" (ghost aliases).

PARTIAL ADDRESS DECODING:

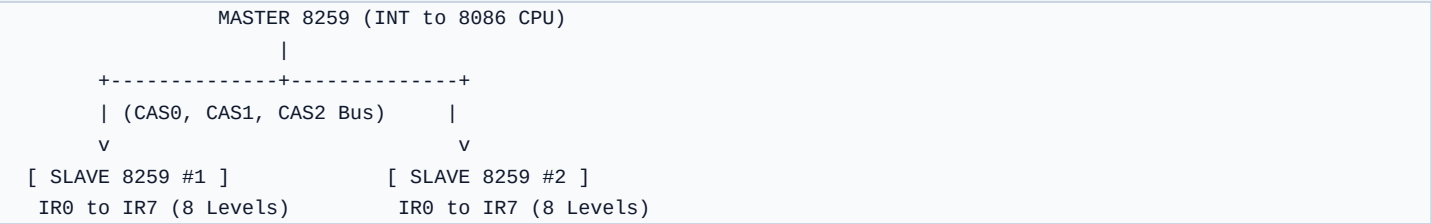
- Only some upper address lines are decoded; unused "don't care" lines (X) are left unconnected.
- Simpler and cheaper hardware (fewer logic gates).
- Causes "Shadow Addresses" (same physical hardware responds to multiple address ranges).

Comparison Table:

Feature	Absolute Decoding	Partial Decoding
Address Line Usage	All address lines (**A_0 - A_19**) ful	Only subset of lines decoded (**A_15-A
Shadow Addresses	**Zero Shadow Addresses** (Strict 1-to	Multiple shadow aliases exist.
Hardware Complexity	High (requires multi-input logic gates	**Low** (simple 74LS138 decoder).
Best Suited For	Large systems, 1 MB full memory maps.	Small microcontroller/embedded systems

2. 8259 Cascaded Mode (Master-Slave Configuration)

To handle more than 8 interrupts, one Master 8259 is connected to up to 8 Slave 8259s via the 3-bit Cascade Bus (CAS_0, CAS_1, CAS_2).



Cascading Operation Flow:

1. Slave 8259 receives interrupt on its IR line and signals Master 8259 via Master's IR pin.
2. Ma

3. Quick Recall Flow

Absolute (All Lines, No Shadows) vs Partial (Don't Cares, Shadow Addresses) | 8259 Cascading: Master-Slave via CAS0-2
-> Up to 64 Interrupts